

Calculating the Number of Particles in Air

1. Research Question

Calculate the number of particles in air using the relationship between the pressure of air and its absolute temperature.

2. Background

2.1 The Pressure Law

The pressure law, also known as Gay-Lussac's law, states that if the volume of a sample of an ideal gas remains constant, its absolute pressure is directly proportional to its absolute temperature in kelvin. This can be expressed as $p \propto T$.

2.2 Air as an Ideal Gas

An ideal gas is a gas whose particles take up negligible space and have no forces between them. They follow the ideal gas law of $pV = nRT$.

Real gasses approximate ideal gasses at high temperatures relative to the critical temperature of the gas, and low pressures relative to the critical pressure of the gas. This is because at relatively high temperatures and low pressures intermolecular forces become negligible due to particles having greater average kinetic energy and being farther apart. The critical point of a gas can be represented in Figure 1 below, which shows the phases of matter at different temperatures and pressures. A real gas approximates an ideal gas when it is well within the bottom right segment of Figure 1 and in the gaseous phase.

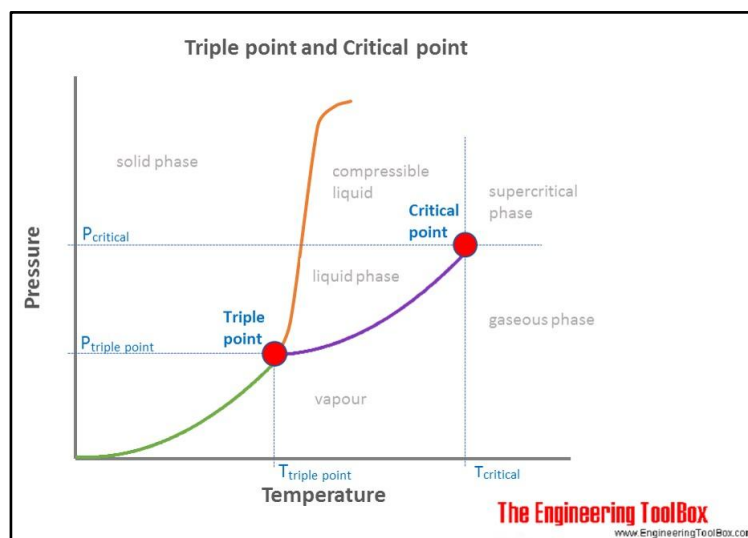


Figure 1: Triple Point and Critical Point (The Engineering Tool Box, 2003)

For Air, the critical temperature is -140.52°C , which is 132.63K and the critical pressure is 3786kPa (The Engineering Tool Box, 2003). In this experiment, the lowest temperature used is 303K , which is relatively high compared to the critical temperature, and the highest pressure obtained is 113.5kPa , which is relatively low compared to the critical pressure, meaning that in this experiment, air will approximate an ideal gas.

2.3 Movement & Interactions of Particles in Air

Since air approximates an ideal gas in this experiment, the behaviour of air particles mimics that of the particles of an ideal gas. This means that each particle has a random direction and kinetic energy and moves freely within the container. Collisions between two particles or a particle and the edge of the container are perfectly elastic, meaning no energy in the air is lost to the surroundings.

2.4 Molar Mass of Air

$$M_{\text{air}} \approx 28.96 \text{ g mol}^{-1} \approx 0.02896 \text{ kg mol}^{-1} \text{ (Unacademy)}$$

2.5 Density of Air at RTP

$$\rho_{\text{air}} \approx 1.293 \text{ kg m}^{-3} \text{ (NASA Earthdata)}$$

2.6 Prediction

Firstly, the pressure law states that for an ideal gas, pressure is directly proportional to its absolute temperature. Hence, graphing pressure with absolute temperature is expected to produce a positive linear relationship.

Next, using the ideal gas law and the values of R and N_a from the IB Physics Data Booklet, I should be able to calculate the number of particles in 59.4cm^3 of air (the volume used in this experiment). The expected value for the number of particles in 59.4cm^3 of air can also be calculated using the molar mass of air and the density of air at RTP:

$$\begin{aligned} m &= 1.293 \times (59.4 \times 10^{-6}) = 7.09857 \times 10^{-5} \text{ kg} \\ n &= \frac{7.09857 \times 10^{-5}}{0.02896} \approx 2.45 \times 10^{-3} \text{ mol} \\ N &= 2.45 \times 10^{-3} \times 6.02 \times 10^{23} \approx 1.47 \times 10^{21} \text{ particles} \end{aligned}$$

3. Methodology

3.1 Variables

Independent variable: the temperature of gas measured using a thermometer at 30°C, 35°C, 40°C, 45°C, 50°C, 55°C, 60°C, 65°C, 70°C, 75°C

Dependent variable: the pressure of gas measured using a pressure gauge.

<u>Controlled variables</u>	Value	Why it is controlled
Gas.	Air.	To calculate the number of particles in a gas, I need to use the same gas, in this case, air.
Time with water at each certain temperature before measuring pressure.	10 seconds.	To allow time for heat to transfer to the water and to the air inside of the conical flask.
Stirring the water.	Constantly.	To ensure that heat is spread evenly within the flask. This means that the thermometer will record an accurate temperature of all of the water in the flask, not just a hot or cold area.

3.2 Apparatus

- 250cm³ beaker
- 50cm³ conical flask (must fit inside of beaker) found to have a volume of $59.4 \pm 0.1 \text{ cm}^3$
- 100cm³ measuring cylinder
- Bung with a hole and tube for pressure gauge connection (airtight)
- Stand
- Clamp
- Heat proof mat
- Tripod
- Mesh square
- Bunsen burner
- Pressure gauge (digital)
- Thermometer
- Stirring rod

3.3 Diagrams

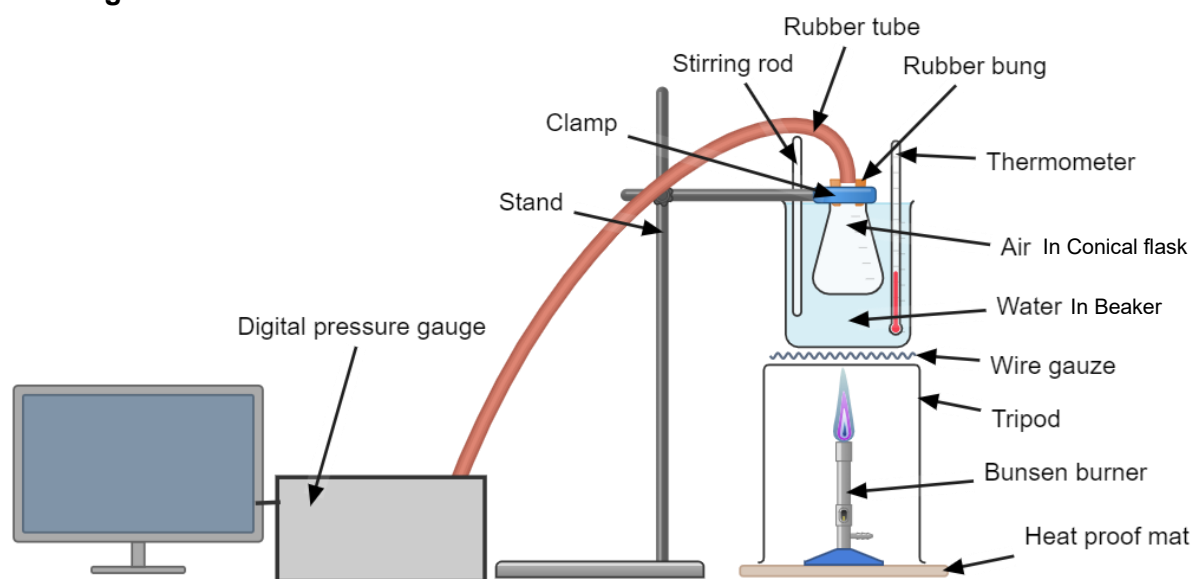


Figure 2: Diagram of Apparatus

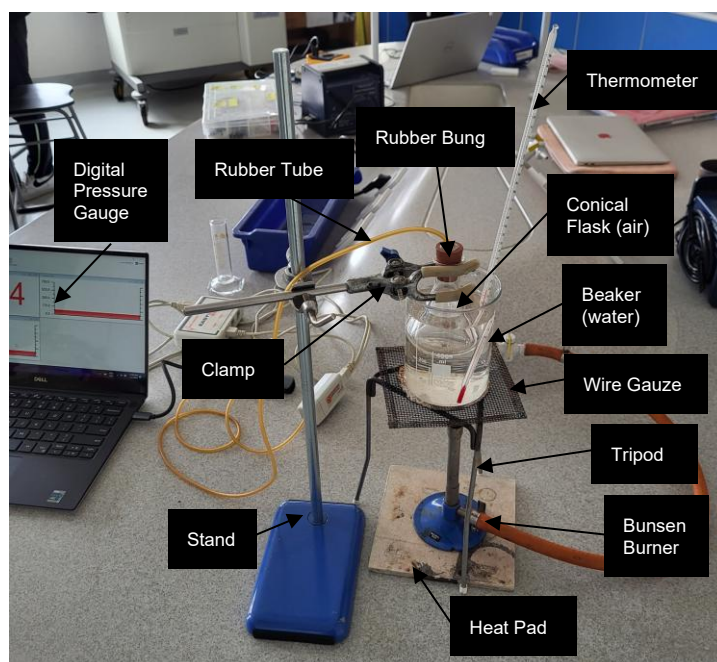


Figure 3: Photo of the Apparatus

3.4 Method

1. Place the bung inside of the conical flask to the point where it is airtight and draw a line on the flask where the bottom of the bung ends.
2. Measure the volume of the conical flask up to the drawn line by filling the flask with water using the measuring cylinder
3. Pour out the water and set up the apparatus
4. Heat the water in the beaker using the Bunsen burner to 30°C measured using the thermometer
5. Measure the pressure of the air in the conical flask using the pressure gauge
6. Repeat steps 4 and 5 for 35°C, 40°C, 45°C, 50°C, 55°C, 60°C, 65°C, 70°C, 75°C
7. Repeat steps 4 to 6, four more times

3.5 Safety

Safety concern	Precautions taken
Dealing with heat and heat sources (i.e. Bunsen burner).	Not wearing loose-fitting clothes or long hair. Not touching anything which was heated until it cools off. Having a fire extinguisher nearby in case of an emergency.
Glassware breaking.	Safety goggles. Apparatus set up away from the edges of the table. Tripping hazards (e.g. chairs and backpacks) moved during the practical.

This investigation poses no significant ethical or environmental issues.

3.6 Preliminary

After conducting preliminary trials, I found that my apparatus was not fully airtight. First I tried to remedy this by using white-tac to seal the connection between the rubber tube and bung, which was ultimately unsuccessful. During the final experiment, I opted for using hot glue to seal the bung to the connecting tube, which provided a sufficiently airtight seal.

4. Analysis

4.1 Raw Data Tables

Temperature (°C) ± 0.5	Pressure (kPa) ± 0.1				
	1	2	3	4	5
30	101.8	101.6	102.0	101.9	101.8
35	104.0	102.4	103.3	103.1	103.3
40	105.2	103.8	104.9	104.5	104.4
45	106.8	104.5	106.1	105.6	106.2
50	107.6	106.1	107.0	106.8	107.1
55	109.4	107.3	108.4	108.2	107.9
60	110.7	108.7	109.8	109.4	109.8
65	111.9	110.4	111.0	110.7	110.9
70	112.6	111.6	112.4	111.9	112.1
75	114.2	112.4	113.5	113.1	113.3

4.2 Processed Data Table

T (K) ± 0.5	Average p (kPa) ± 0.1	Δp (kPa)
303	101.8	0.2
308	103.2	0.8
313	104.6	0.7
318	105.8	1.2
323	106.9	0.8
328	108.2	1.1
333	109.7	1.0
338	111.0	0.8
343	112.1	0.5
348	113.3	0.9

4.3 Graph

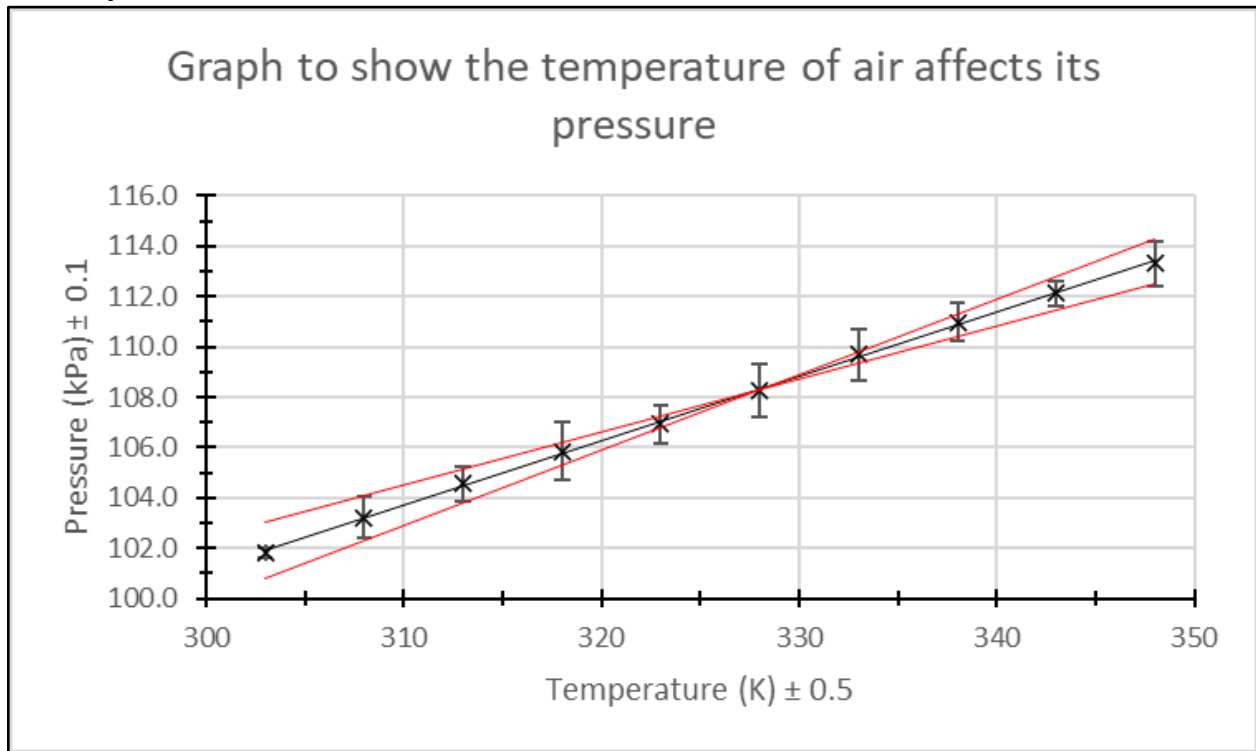


Figure 4: Graph of Results

Line of best fit: $y = 0.2552x + 24.539$

Lines of worst fit: $y = 0.3x + 9.9$ and $y = 0.21x + 39.4$

4.4 Sample Calculations

Calculating the temperature in K:

$$\text{Temperature (K)} = \text{Temperature (}^{\circ}\text{C)} + 273$$

$$T_1 \text{ (K)} = 30 + 273 = 303 \text{ K}$$

Calculating average pressure:

$$p_{avg} = \frac{p_1 + p_2 + p_3 + p_4 + p_5}{5}$$

$$p_{avg} = \frac{101.8 + 101.6 + 102.0 + 101.9 + 101.8}{5} = 101.82 \approx 101.8 \text{ kPa}$$

Calculating the uncertainty of average pressure:

$$\Delta p = \frac{\text{largest } p - \text{smallest } p}{2}$$

$$\Delta p = \frac{102.0 - 101.6}{2} = 0.2 \text{ kPa}$$

Calculating the uncertainty of the line of best fit (gradient):

$$\Delta \text{gradient} = (\text{max gradient} - \text{min gradient})/2$$

$$\Delta \text{gradient} = (0.3 - 0.21)/2 = 0.045 \text{ kPa } K^{-1}$$

$$\Delta \text{gradient} = \frac{0.045}{0.2552} \times 100\% \approx 17.6\%$$

Calculating number of moles in 59.4cm³ of air:

$$pV = nRT$$

$$n = \frac{pV}{RT}$$

$$n = (\text{gradient}/1000) \times \frac{V}{R}$$

$$n = (0.2552/1000) \times \frac{59.4}{8.31} \approx 1.82 \times 10^{-3} \text{ mol}$$

Calculating the uncertainty in the number of moles in 59.4cm³ of air:

$$\Delta n = n \left(\frac{\Delta \text{gradient}}{\text{gradient}} + \frac{\Delta V}{V} \right)$$

$$\Delta n = (1.82 \times 10^{-3}) \left(\frac{0.045}{0.2552} + \frac{0.1}{59.4} \right) \approx 3.2 \times 10^{-4} \text{ mol}$$

Calculating the number of particles in 59.4cm³ of air:

$$n = \frac{N}{N_a}$$

$$N = nN_a$$

$$N = 1.82 \times 10^{-3} \times 6.02 \times 10^{23} \approx 1.10 \times 10^{21} \text{ particles}$$

Calculating the uncertainty in the number of particles in 59.4cm³ of air:

$$\Delta N (\%) = \Delta n (\%) \approx 17.8\%$$

$$\Delta N = (1.10 \times 10^{21})(0.178) \approx 1.95 \times 10^{20} \text{ particles}$$

4.5 Conclusion

In conclusion, the graph shows a positive linear relationship between the pressure and absolute temperature of the air, which matches the prediction. The equation of the line of best fit is $y = 0.2552x + 24.539$, where y represents pressure (kPa) and x represents absolute temperature (K). This equation was used to calculate an experimental result of 1.10×10^{21} particles with an uncertainty of 17.8%.

According to Gay-Lussac's law, $p \propto T$, so the line should cross the origin, meaning the equation should have a constant term of 0 instead of 24.539. This can be explained most simply by understanding that as the temperature nears absolute zero, air will liquefy and solidify, thus no longer approximating an ideal gas, which would always remain in a gaseous state.

Changing y to represent pressure (Pa), the gradient of the equation is multiplied by 1000, giving a gradient of 255.2 with an uncertainty (calculated above) of 17.6%. According to the ideal gas law $pV = nRT$, thus $p = \frac{nR}{V}T$ meaning the gradient of 255.2 represents $\frac{nR}{V}$, which was used to calculate the experimental result for the number of moles and ultimately the number of particles.

The expected result of 1.47×10^{21} particles is slightly outside of the uncertainty range of the experimental result of $1.10 \times 10^{21} \pm 1.99 \times 10^{20}$ particles. This shows that there are systematic errors in the experiment that cause this uncertainty, which I discuss in the next section. The uncertainty value of 17.8% is less than the average relative uncertainty of 24.3% found by Hadiningrum, K. & Muldani, R. (2018), using a similar Gay-Lussac's law investigation to find the number of moles of a volume of air with similar apparatus. Finding the number of particles in my investigation is equivalent to finding the number of moles, as only a simple calculation using Avogadro's constant was used to convert from moles to particles, so this investigation by Hadiningrum, K. & Muldani, R. can be compared directly to my own.

5. Evaluation

5.1 Strengths of Investigation

Strength	Why it is a strength
Apparatus was airtight. Using hot glue as an air-tight seal worked well.	No air particles were able to enter or exit the system, meaning that the number of moles of air was held constant. This is important as a varying number of moles would result in Gay-Lussac's law not applying.
Water was constantly stirred and time was left for heat to spread.	This allowed for more accurate measurements of temperature, as heat was spread evenly within the beaker and the thermometer had sufficient time to reach and stop at the correct value.

5.2 Limitations of Investigation & Improvements

Limitation (Systematic Error)	How it affects results & significance	Improvement
Volume of air in the connecting tube was not counted.	The recorded volume of the system being heated was less than it should have been. This is highly significant as the volume in the tube was relatively large.	Fill connecting components with water as well as the conical flask to measure the volume correctly.
The pressure gauge jumped values by 0.2 or 0.3kPa at a time.	The uncertainty of the pressure gauge used for calculations was inaccurate. This is slightly significant, as it impacted pressure results by a small amount.	Use a more accurate pressure gauge, which does not skip any values.
Heat transfer from water to conical flask to air.	The heat was transferred from the water to the air through the conical flask through conduction. When heat is transferred energy is lost to the surroundings, so the temperature of the air was likely lower than that of the recorded value of the water. This is slightly significant, as heat loss is often quite high.	Reduce the number of heat transfers needed to heat the air. Either heat air directly (from within the closed system), or heat through only the conical flask (which may need to be replaced with a boiling flask).

(limitations continued on next page)

Limitation (Systematic Error)	How it affects results & significance	Improvement
Heat transfer within the air.	Heat is transferred through convection within the air. This means that the hotter air would travel upwards, in this case towards the top of the conical flask and through the connecting tube. So the air at the end of the tube likely has a higher temperature. This is insignificant as the air is at a high temperature so particles travel quickly spreading heat, and the volume of air used was relatively small.	The connecting tube could be shortened and the volume of gas could be decreased to make this effect even more insignificant as there will be less air for the heat to transfer within.
The time left with the water at a certain temperature allowed heat to escape the system.	The pressure recorded is slightly less than expected. This is slightly significant as a small amount of heat could leave the system.	Insulate the beaker of water and use a lid on it to ensure a minimal amount of heat loss to the surroundings.
Air is not an ideal gas.	The ideal gas equations used to calculate the result only apply to ideal gasses. This is insignificant, as at the temperatures and pressures in this experiment air approximates an ideal gas.	Use higher temperatures, as real gasses approximate ideal gasses at higher temperatures and lower pressures relative to their critical point.
The air used does not have the exact same chemical composition as the expected value.	It would have a slightly lower or higher molar mass, thus it increases the uncertainty of the obtained result by an unknown amount. This is insignificant, as the contents of air samples remain largely the same and there were no significant gasses produced around the area where the sample was taken.	The chemical composition of the air used in the experiment could be found using spectroscopy to better calculate the expected result.
Heat transfer within the water. Not much space for the stirring rod.	The heat was not spread completely evenly in the water. This is insignificant, as the beaker held only a small amount of water so not much stirring was needed for heat to spread evenly.	Use a larger beaker of water to increase the space.

5.3 Expansion of Investigation

The same or similar methods used in this investigation could be used to find the number of moles and the number of particles in any volume of any gas. Using a pure gas would be interesting because it better approximates an ideal gas, as all particles are the same shape.

It would also, however, be interesting to see how conducting the investigation at very low temperatures or very high pressures relative to the critical point of air would affect the results, as air would no longer approximate an ideal gas. This would require special equipment to either cool down the air closer to its boiling point or to withstand very high pressures and hence would require the method to be altered.

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